



(Im)material Streams in the City: Ecologies, Economies, Electrons

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FROM THE GUEST EDITORS



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Thinking through the Materiality of the Digital

Processes of datafication have been advancing at full speed for several years now, reaching all aspects of human life and beyond (Mejias and Couldry, 2019; Nost and Goldstein, 2022). This ever-increasing production of data has been promoted *ad nauseam* by the tech industry as a promise for brighter futures, and public institutions have largely bought into the narrative that more data is key to solving their operational problems (Ashuri, 2025). The geography of this datafication process has emerged as a key object of inquiry over the last couple of decades (Mackinnon et al., 2022). Cities, in particular, have attracted a lot of attention as spaces where large populations will generate more data, and ultimately lead to higher tech firm profits (Barns, 2020).

Meanwhile, scholars have acknowledged that data not only have a geography, but a materiality as well—in sharp contrast to their apparent weightlessness and ephemerality. This materiality can take multiple forms and has thus been documented and analyzed in various ways and from different perspectives. The most evident aspect of this materiality has to do with the spatial imprint of the infrastructure that allows for the circulation of information: copper wires and coaxial cables, servers and Internet exchange points (Castells, 1989; Batty, 1997). Scholars interested in the development of large technical networks (such as electricity, water, or network heating) were quick to embark on this research agenda and started examining the territorial reconfigurations that followed the extension of telecommunication infrastructures (Graham and Marvin, 1996; Dupuy, 2002), while others called for these networks to be integrated to the scope of cities' planning efforts (Shiode, 2000). These reflections helped advance our understanding of how digital infrastructures reshape or reinforce existing socio-spatial inequities, with authors extending this line of thinking to analyze how marginalized groups can reclaim Internet infrastructures and use them for (digital) self-determination (Sandvig, 2013; McMahon, 2014).

While scholarship on the infrastructures of the digital mostly revolved around the deployment of telecommunication networks, a more delineated research object emerged as a new way to interrogate digital materiality: data centers. With their multiplication in number and in scale, data centers have become an important focal point to interrogate how digital activities consume resources (Hogan and Vonderau, 2019; Levenda and Mahmoudi, 2019; Ortar et al., 2023), reshape territories and existing urban fabric (Brodie and Velkova, 2021; Taylor, 2023), and transform labor conditions for the workers who operate and maintain digital infrastructures (Carnino and Marquet, 2019; Monserrate, 2023). With their environmental footprint expanding—in 2022, data centers accounted for about 2 percent of all global energy use, and the amount of raw energy consumed by data centers doubles roughly every four to eight years (International Energy Agency, 2024)—interest is quickly growing. This body of work now constitutes an important field that draws from numerous disciplines

and domains, including (but not limited to) media studies, geography, science and technology studies, anthropology, and history (Edwards et al., 2024; Pronovost et al., 2024). It has proven to be a window into the material manifestation of “the cloud,” with local or regional implications where the infrastructure is deployed (e.g., the use of resources to power it up and cool it), as well as global ones—here, consider the extractive practices and industrial processes that are necessary to produce the electronic components that populate data centers, only to later be discarded as e-waste (Bridges et al., 2023).

The digital has always been material, but the advent of large language models and more powerful artificial intelligence (AI) technologies seem to be calling for a shift in how we think about this materiality. To the best of our ability to measure it, training GPT-3, the large language model behind a dated version of ChatGPT, required its parent company OpenAI to use enough fresh clearwater to fill up a nuclear reactor’s cooling tower (i.e., over 840,000 L)—bearing in mind that subsequent versions of the model rely on larger sets of data parameters and likely gobbled up more resources when they were trained (DeGeurin, 2023). Estimating carbon emissions of AI is difficult, but some estimates show that training a transformer model such as Llama3.1 (Meta’s flagship AI model) emits as much carbon as 10,000 vehicular round trips from Los Angeles to New York City (Han et al., 2024), and the cumulative use of such models emits as much as 25 times this amount each year (Chien et al., 2023). How much electricity is consumed to answer a question using a large language model, compared to obtaining the answer using a “conventional” search engine? “Four to five times the energy” (Crawford, 2024)? But then again, a Goldman Sachs report that has widely circulated in the media says ten times (Goldman Sachs, 2024). Do we base these estimates on how much energy a given prompt needs to be processed, or do we take into account the initial, highly energy-intensive training of the model? Not only do industry leaders intentionally obfuscate the ecological impacts of training and maintaining their models, leading to many assessments based on off-hand comments and best-guess estimates (Luccioni et al., 2025), but at the same time, the material implications of emerging technologies seem to require shifts in measurements and scholarly engagement. We are faced with an “avalanche of numbers,” but while Hacking (2016) used this phrase to designate the rise of statistics as an instrument of government, this multiplicity of numbers increases uncertainty and makes it harder to govern.

Beyond hardware and the resources it consumes, digital materiality extends to human bodies: as Hayles (1993: 122) puts it, informatics also include “habits of posture, eye focus, hand motions and neural connections that are reconfiguring the human body in conjunction with information technologies.” The Latin roots of the word “digital” is telling in this regard, because *digitalis* refers not only to small numbers (“digits”) but also to “fingers”—a key element in everyday human–machine interactions. But here again, generative AI renews how we think about the material. For example, a recent study showed that using generative AI technologies for writing produces lower brain activity levels than writing without them (Kosmyrna et al., 2025), leading to heated debate across social media about the merits of the study and its broad scholarly and pedagogical implications: what the study provoked is a move beyond the technology-as-tool or technology-as-society debate into the realm of measuring and theorizing the materiality of technology and its use. Again, generative AI widens and deepens the material implications of digital technologies, but these implications are more difficult to ascertain, leading to more conflictual public debate.

How are we to make sense of these environmental impacts, that are simultaneously difficult to ascertain and diffuse in the theoretical frameworks that should be brought to bear upon them? Does this call for new ways of thinking about the flows and circulations of resources in urban ecologies?

Digital and Material Circulations

The collection of articles in this issue of *JUT* includes five contributions, with research sites spread across several continents (Africa, Europe, Latin America, and Europe). We believe that this is important, because of the global reach and interconnectedness of digitalization processes—including their material implications (Arboleda, 2020). Equally important was the need to produce more empirical investigations of digitalization processes in the global South (Datta, 2018), and to theorize digital/material circulations from peripheral spaces (Lawhon and Truelove, 2020; Palat Narayanan, 2021).

Each article approaches the dialectic between digital and material circulations through a unique perspective. Andrea Pollio's article, "Of Bloatware and Spreadsheets: Nairobi, Chinese Phones, and the Limits of Data Coloniality," offers a compelling illustration of the material dimension of datafication through an analysis of Chinese cellphone deployment in Nairobi, Kenya. Bloatware and spreadsheets—which form the title of this work—are two material artifacts of datafication that allow for an analysis of the complex relationship between the hardware and data circulations. These circulations create new geographies, as Nairobi emerges as a form of "frontier market" not characterized by the late diffusion of new technologies (in this case, Chinese low-cost cellphones) but by trials and experimentation, transformation and rectifications. These phones' user data are sent to Chinese parent companies and eventually reshape the market globally, while the development of a local market for these devices reshapes Nairobi's urban fabric locally.

Meanwhile, Julia Frotey, in "Connected Charging Stations to Govern Individual Mobility: The Contribution of Digitalized Electronic Mobility Networks to Sustainable Mobility Policies in France," approaches the question of circulations through the lens of urban mobility, and more specifically through an inquiry into the politics of data generated by electric vehicles' charging stations. She analyzes the mutual constitution of the digital and material infrastructures that subtend electric mobility, emphasizing how producing and displaying data were key to making charging stations functional for end-users, and how the use of algorithms increasingly dictates their deployment, expansion, relocation, or decommission. We are thus reminded of how the circulation of data conditions the expansion of infrastructures, and in turn, the movement of people across urban space.

"Divergent Futures in a Damaged Territory: The Rise of Data Centers and Water Conflicts in Santiago de Chile," by Martin Tironi and Camila Alborno addresses the materiality of the digital by examining a Google data center project in Cerrillos, a municipality that forms part of Santiago de Chile, whose construction would involve a massive diversion of water resources. Here, the authors point not only to the metabolic footprint of digital infrastructures, but they also highlight that the reworkings of flows and circulations are contingent on the mobilization of a discursive apparatus. The article underlines the confrontation between an "eco-modernist" narrative leveraged by proponents of the data center and an alternative worldview put forward by activists who sought to "re-materialize" this infrastructure and emphasize its hydric consumption in a context of water scarcity. Here, community activists managed to get a ruling preventing Google from using water to cool its servers, showing that tech giants' worldviews are not all-powerful, and that alternative visions of the future can be conjured to realign digital and material circulations in a more equitable way.

While the Chilean case study shows that circulations of matter and data can represent important points of contention, Erik Bordeleau and Nathalie Casemajor show how activists can use them to their advantage in their article "Interspecies Cyber-Governance: Bee DAO and the Artistic Imaginaries of Blockchain for Planetary Regeneration." This argument is built upon the analysis of a unique experiment that blends urban activism, artwork, beekeeping, and blockchain technology. The Berlin-based project involves sensors that provide data

on the condition of beehives (e.g., humidity, temperature, weight), which feed into a decentralized autonomous organization aimed at reflecting the health of the bee population. This assemblage of data and cryptographic tools, human and non-human living entities, buildings and beehives, is designed to make a statement about the organization of collective governance around, and in the name of, natural ecosystems—in this case, the bees.

In the last article “All That is Urban Melts into Data: Circulation of Matter, Energy, and Data in the Digital City,” Mouton and Burns draw upon the previous articles to put forward a theoretical contribution allowing us to think through the intertwining of digital and material circulations. It does so by establishing closer dialogue between critical data studies (CDS) and urban political ecology (UPE). For UPE, the challenge is to integrate the movement of data to the framework of urban metabolism, emphasizing the related use and transformation of material resources that allow their motion, computation, combination, or storage. In turn, a conversation with UPE provides CDS with an opportunity to “urbanize” reflections on digitalization, where cities are not a mere backdrop but an important actor for processes that involve the circulation, conversion, and accumulation of data.

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